

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1263

Country Club 3 100 kV
April 02, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1263. The Interconnection Customer proposes a 104.9 MW solar photovoltaic facility interconnecting at the Country Club 3 100 kV substation in the Indiana area. The request was received on July 31, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 384 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,660,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Montpelier Electric Generating Sta - Coso 7 100kV	Muskogee - Montpelier Electric Generating Station 500kV L	125.7	156	14.7
Sabanera - Murdock 230kV Line	Henry County - E Methuen 500kV	101.5	731	8.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Montpelier Electric Generating Sta - Coso 7 100kV	Rebuild line with higher-capacity conductor	\$19,343,000	33% (\$6,433,000)
Sabanera - Murdock 230kV	Reconductor line	\$18,442,000	1% (\$227,000)
	Total allocated to project		\$6,660,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.